Question 5.

START

function displayMenu()

Print 1. Register member

Print 2. Display Available books

Print 3. Borrow a book

Print 4. return a book

Print 5. display member record

Print 6. Exit

Print Enter your choice

End function.

function registerMember (name, studentID, memberType)

name ← input.

student ID ← input

Repeat

memberType ← input (1 = student, 2 = Faculty)

If memberType != 1 AND memberType != 2

print invalid type try again

End if

until memberType == 1 OR memberType == 2.

Print registration successful

End function.

function displayBooks (bookIDs, bookTitles, availableCopies, totalBooks)

Print "ID, Title, Copies"

for i = 0 to totalBooks - 1

```
        Print bookIDs[i], bookTitles[i], availableCopies[i]
    Endfor.
End function
Function borrowBooks(bookIDs, bookTitles, availableCopies,
                     totalBooks, borrowedBookIDs, borrowed(
                     memberType)
If borrowedCount >= 3
    print "You already have 3 books"
    Return
End if
id ← input
setindex ← -1
for i=0 to totalBook-1
    if bookIDs[i]==id
        set index ← i
    End if
End for.
IF index==-1
    print invalid bookid
    return
End if.
if availableCopies[index]==0
    print no available copies
    return
Endif
```

```
days ← input
if days<=0
    print number of days must be +ve
    return
endif
if memberType == 1
    Set allowedDays ← 14
else
    Set allowedDays ← 30
End if
if days > allowedDays
    print limit → allowedDays, fine applies after that
End if
Set borrowedBookIDs [borrowedCount] ← id
Set borrowedCount ← borrowedCount + 1
Set availableCopies[index] ← availableCopies[index] - 1
Print
    you borrowed bookTitles[index]
End Function
function returnBooks (BookIDs, availableCopies, totalBooks,
                      borrowedBookIDs, borrowedCount, memberType,
                      totalFine)
Id ← input
Set bookIndex ← -1
set borrowPos ← -1
```

```
For i=0 to borrowedCount-1
    if borrowedBookIDs[i] == id
        Set borrowPos = i
    End if
End for
If borrowPos == -1
    print "you have not borrowed this book"
    return 0
End if
days ← input
if days <= 0
    print number of days must be positive
    return 0
End if
For i=0 to totalBooks -1
    if bookIDs[i] == id
        Set bookIndex = i
    End if
End for.
Set availableCopies[bookIndex] = availableCopies[bookIndex]
If memberType == 1
    Set allowedDays ← 14
    Set rate ← 20
Else
```

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```
        set allowedDays ← 30
        set rate ← 10
    Endif
    Set fine ← 0
    if days > allowedDays
        set extraDays ← days - allowedDays
        set fine ← extraDays * rate
        print late return, extraDays, fine
        set totalfine ← totalfine + fine.
        membership code ← input.
        If membership code == "LIB2026"
            set fine ← fine/2
            print 50% waiver applied fine now; fine
        Else
            print wrong code, no waiver, fine; fine
        End if.
    Else
        Print returned on time, no fine
    Endif
    for i: borrowPos to borrowedCount-2
        Set borrowedBookIDs[i] = borrowedBookIDs[i+1].
    End for
    ← Set borrowedCount = borrowedCount-1
    ← Return fine
End function
```

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function displayMemberRecord (name, studentID, memberType, borrowedBookIDs, borrowedCount, totalFine, fineAfterWaiver)

print name, studentID

If memberType == 1

print memberType: Student.

else

print memberType: Faculty.

Endif.

Print current borrowed books.

if borrowedCount == 0

print none

Else

for i=0 to borrowedCount-1

print borrowedBookIDs[i]

Endfor.

Endif.

print total fine before and after waiver.

End function.

Main

Declare bookIDs[4] = {b1, b2, b3, b4}

Declare bookTitles[4] = {"Programming in C", "Data structures", "Database Systems", "Computer Networks"}

Declare availableCopies[4] = {5, 3, 0, 4}

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```
Set totalBooks ← 4
setShudentID ← 0
Set membeType ← 0
Declare name, borrowedBookIDs[3]
set borrowed count ← 0
set totalfine ← 0
set fineAfterWaiver ← 0
Set registered ← 0
Repeat
    Call displayMenu()
    choice ← input
    Switch choice
        Case 1:
            If registered == 1
                print already registered
            else
                call registerMember(name, shudtID,
                                    memberType)
                Set registered ← 1
            End if
            Break
        Case 2:
            Call displayBooks (bookIDs, bookTitles,
            available copies, totalBooks)
            Break
```

Case 3:

if registered == 0

print please register first.

else

call borrowBook(bookIDs, bookTitle, availableCopies, totalBooks, borrowedBookIDs, borrowedCount, memberType)

End if.

Break

Case 4:

if registered == 0

print please register first

Else

set fineAfterWaiver = fineAfterWaiver + call returnBook(bookIDs, availableCopies, totalBooks, borrowedBook borrowedCount, memberType, totalFine)

End if

Break

Case 5:

if registered == 0

print please register first

```
        else
            call displayMemberRecord(name, shadedID, memberType,
                borrowedBookIDs, borrowedCnt, totalFine,
                fineAfterWaiver)
        End if
        Break
    Case 6:
        print Goodbye
        Break;
    Default;
        print invalid choice
End switch
until choice == 6
STOP
```

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Input:

- Menu Choice
- Name
- Student ID
- Membership Type
- Book ID
- Number of days.
- Membership Code

Output

- display menu
- Registration & display registration confirmation or invalid membership type
- Book inventory
- Borrowing
- Returning
- Member Records
- Exit

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